Erick Barraza

IT projects — network tooling, subnet auditing, and hands-on Windows 11 administration

GITHUB github.com/ejbarraza

1 Network Port Scanner

OPEN SOURCE

github.com/ejbarraza/port-scanner

A dependency-free Python TCP port scanner built only on the standard library: threaded scanning, CIDR target support, quick / full / custom port profiles, banner grabbing and reverse DNS, with reports in text, JSON, CSV, and HTML.

- Backed by 25 pytest tests, with GitHub Actions CI running the suite across Python 3.10–3.12.
- Field-tested on a live subnet: found and fixed two real defects — a reverse-DNS stall on dead hosts and a Ctrl+C path that drained the probe queue instead of stopping — each covered by a regression test.

2 Virtual Lab Network Audit

HOMELAB

Findings: github.com/ejbarraza/port-scanner/tree/main/docs/lab-findings

Applied the scanner to a full VirtualBox NAT subnet (10.0.2.0/24, 254 hosts) from a Windows 11 VM, completing a sweep of every address in under five minutes.

- Mapped 7 open ports across the subnet, including SMB (445) and MSRPC (135), and documented what each exposed service is.
- Recorded per-service remediation notes — whether each port should be exposed and how to close or restrict it.

3 Windows 11 System Administration Lab

HOMELAB · 21 LABS

Built and administered a Windows 11 (64-bit) VirtualBox VM — 80 GB GPT disk, NAT networking — across 21 documented, screenshot-verified hands-on labs.

- **Networking:** TCP/IP configuration, network-adapter identification and enable/disable, active-connection and Wi-Fi inspection, connectivity testing (ping, 0% packet loss), and break/fix troubleshooting.
- **Storage:** disk initialization and volume creation in Disk Management and DiskPart, including building and attaching a 2 GB virtual hard disk (VHD).
- **Devices & hardware:** Device Manager driver verification, printer installation, and input/output device configuration (keyboard, mouse, display, audio).

TECHNOLOGIES EXERCISED

Python (stdlib) · pytest · GitHub Actions CI · Windows 11 · VirtualBox · TCP/IP & DNS · DiskPart · Device Manager

Full lab documentation and scanner source are on GitHub; happy to walk through any of it in an interview.